

We continue by finishing the discussion on zero dynamics attacks. We've seen the other day the unmatched initial conditions, so with $x_{k_0}^a = 0 \neq z_{k_0}$.

We've seen how, starting from this initial condition in which they're different, we can compute the distance with $e_k = x_k^a - z_k$, and we'll have that this will go to zero with time. We've computed the evolution of e_k , and it's given by $e_{k+1} = Ae_k$. We've also seen the evolution of $z_{k+1} = (A + BF)z_k$ with $a_k = Fz_k$, and $y_k^a = Cx_k^a = Ce_k + \underbrace{Cz_k}_{=0} = Ce_k$.

We want $e_k \rightarrow 0$ as time passes. To do this, we write $e_k = A^k e_{k_0}$. Let's say $e_{k_0} = 0$. Then we said that we compute eigenvalues $(\lambda_1, \dots, \lambda_n)$ of A , and the corresponding eigenvectors $(u_1, \dots, u_n)$. We assume eigenvalues are n real and distinct.

We write $e_k = A^k e_0 = A^k (x_0^a - z_0) = -A^k z_0$. We write z_0 (to be decided by the attacker) in the new coordinate system.

$$\begin{aligned} z_0 &= c_1 u_1 + c_2 u_2 + \dots + c_n u_n \longrightarrow e_k = -A^k z_0 \\ z_0 &= -c_1 \lambda_1^k u_1 - \dots - c_n \lambda_n^k u_n \end{aligned}$$

To have $e_k \rightarrow 0$ the attacker must take $c_i = 0 \forall i$ such that $|\lambda_i| > 1$ (the corresponding eigenvalues are unstable).

This means the initial state is orthogonal with respect to the coordinate axes corresponding to unstable modes of A .

If above criterion for c_i is respected, we have an initial impact $\|y_k^a\|$, but then we reach a point in which we have the matched (ideal case).

The norm of y_k^a depends on z_0 . Thus, the energy of y_k^a can be made small by taking small $\|z_0\|$.

So far we've discussed on detectability, in two cases:

1. In matched conditions (ideal), and this is perfectly undetectable (as we have no impact on the measured output).
2. Unmatched conditions (real), it's practically undetectable as the impact y_k^a can be made small.

Now, let's analyse the impact of the attack. We'll analyse the variable

$$x_k^a \stackrel{\Delta}{=} x_k - \bar{x}_k.$$

Ideally, the attacker wants to have $\|x_k^a\| \rightarrow \infty$, so that at some point x_k will leave the safety set.

Suppose that initially we're not in matched conditions, but we know that eventually we can recover matched conditions. We have:

$$\begin{cases} x_{k+1}^a = Ax_k^a + Ba_k \\ z_{k+1} = (A + BF)z_k \\ a = Fz \\ y_k^a = Cx_k^a \end{cases}$$

We thus have:

$$z_{k+1} = (A + BF)z_k$$

The solution is given by:

$$z_k = (A + BF)^k z_0$$

So far, z_0 was chosen with this rule: $c_i = 0 \forall i \mid |\lambda_i| \geq 0$ of A matrix.

How can we ensure now that $\|z_k\| \rightarrow \infty$? We compute the eigenvalues and associated eigenvectors of $(A + BF)$, and we call them $\bar{\lambda}_i$ and $\bar{u}_i$. Then, we write $z_0 = \bar{c}_1 \bar{u}_1 + \dots + \bar{c}_n \bar{u}_n$. Finally, take $\bar{c}_i \neq 0$ for at least one i such that $|\lambda_i| > 1$. In this way, z will explode, and in matched conditions x will do the same (diverging).

So, here are the attack rules:

- F to be selected to make a given $\mathcal{V}^* \subseteq \text{Ker}(C)$ $(A + BF)$ -invariant. This means that z_k will forever be in $\mathcal{V}^*$.
- In matched conditions, $y_k^a = Cx_k^a = Cz_k = 0$, making the attack perfectly undetectable.
- In reality, first we recover asymptotically matched conditions. We select $z_0 = \sum_{i=1}^n c_i u_i$, with u_i eigenvectors of A . The rule to recover matched conditions is to have $c_i = 0 \forall i$ such that $|\lambda_i| > 1$.
- Impact rule, given $z_0 = \sum_{i=1}^n \bar{c}_i \bar{u}_i$. And the rule is $\bar{c}_i \neq 0$ for at least one i such that $|\bar{\lambda}_i| \geq 0$.
- F chosen to have unstable $(A + BF)$. In this case, we're generating an unstable dynamic, which is not visible to the sensors.

What are the protection strategies? In practice, they're undetectable. And to actually achieve detectability, we need to change something in the system. One strategy is to change A , such that $A \rightarrow (A + \Delta A)$. Here we're physically changing the system, and it's actually easy to do in practice, as systems are highly

redundant. This is also the concept of the moving target. Another possibility is to change B , having $(B + \Delta B)$. For example, through the adding of actuators (or the removal). Finally, we can change C , for example by adding or removing sensors.

What are the effects of these options? When the attacker selects the original z_0 , he is sure it will remain forever in $\mathcal{V}^*$. Eventually, also the trajectories of x_k^a will go towards z_k . After implementing the corrections, the equations of the system will change. The attacker will not know that we're changing the system.

How to select, in a rational way, ΔA , ΔB and ΔC ? How do we implement protections?

If we look at output equations, we have:

$$\begin{pmatrix} y^a 1 \\ \vdots \\ y_{n_y}^a \end{pmatrix} = \begin{pmatrix} * & \dots & * \\ \vdots & & \vdots \\ * & \dots & * \end{pmatrix} \begin{pmatrix} x_1^a \\ \vdots \\ x_{n_x}^a \end{pmatrix}$$

If we remove sensors, we'll have a shorter y vector. We want to decide C to have a smaller Kernel set, having the smallest possible intersection between the kernel of the new matrix and vector $\mathcal{V}^*$.

When we remove sensors, we start from the C matrix, and "cut" some columns. This is not a good action, as we'll have $Ker(C) \subseteq Ker(\bar{C})$.

The good action is to add sensors.

As for the modifications of B , we need to change the number of actuators.

Removing them is in general not recommended. Adding actuators is also not effective.

We have a scaling mechanism to solve the problem, multiplying the control and the divide it with the same value (a sort of encryption and decryption mechanism). This will have an impact of $(A + BF)$ –invariance.

A , for the same reasons, has an impact on the alignment of x_k^a and z_k trajectories.

Covert Attacks

We have a process P , which takes the input u and disturbances w . We also have the measured output y , which is summed to noise n . The result goes into a communication network, and it will emerge from the communication network. It will go to the controller C , which will select u such that y will be close to y_{ref} .

This is in nominal conditions, as $y_m \approx y$, and the computed control u_c is equal to the applied control u .

In zero dynamics attacks, we're controlling the measured output in the system. With that, we have large impact on x and no impact on y .

Excursus: The resources of ZDA are (A, B, C) , so complete model knowledge, as well as disruption resources on u . We don't need disclosure resources as it's an open-loop attack.

In covert attacks, in the other hand, we don't want to disrupt x variable, rather change the output of the system, and to remain cover they inject both something in the command channel, as well as the measurement channel.